

PRE-ANALYSIS PLAN

Realigning climate-forcing players?

Power Resources Mobilisation pathways in the Just Transition

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1. Introduction: CFA stranding and its political implications

Climate change and decarbonization imperatives have led governments across the world to adopt policies aimed at reducing the economic value of 'climate-forcing assets' (CFA), such as coal mines, fossil fuel-powered vehicles, and aircraft reliant on traditional jet fuel.

Decarbonisation policies, however, rarely affect climate-forcing asset holders alone (Colgan, Green, and Hale, 2021).

More often than not, CFA 'stranding' comes at a cost for the *social fabric* of established communities, starting with the fragmentation of social structures at the local level to the contestation of democratic structures by anti-climate parties, as observed in the US, Argentina, or the Netherlands.

From a political economy perspective, this context raises two major questions:

- 1) Can asset-stranding policies be designed in a way that enhances (rather than undermines) local community ties?
- 2) To what extent can political intermediaries (such as trade unions) embedded in CFA structures initiate, cope with, and react to large-scale shifts in relative valuations?

2. Existing contributions and research gap: The role of just transition policy support

2.1. The political backlash of 'climate-forcing asset holders'

Climate change and decarbonization policies have an obvious distributive impact. As scholars have shown (Colgan, Green, and Hale, 2021), climate politics across the globe can be essentially portrayed as a struggle between two groups: To the extent that they are negatively impacted by climate change, 'climate vulnerable asset holders' (CVA) generally benefit from climate policies. On the other hand, however, exists a group of 'climate-forcing asset holders' (CFA), which can be broadly defined as those industries, workers, and citizens, set to be negatively affected by decarbonization policies.

While decarbonization policies may rely on the support of CVAs, the political disturbance potential of CFA action cannot be underestimated. Consumers are strongly responsive to price signals from decarbonization policies, as showed in the case of energy prices in the Netherlands (Voeten, 2024) or the ban on polluting cars in Milan (Colontone et al., 2023) – which provides CFAs with a strong status quo bias. Besides, recent events have shown that policy resistance and political mobilization by CVAs increasingly translate in anti-climate parties winning major national elections (e.g. US election in 2024, Argentine election in 2023, the Netherlands in 2023, etc.).

In sum, climate change policies won't be adopted without a certain degree of realignment of climate forcing asset holders.

This context raises the question as to the conditions under which CFAs **may be realigned to consent to climate policies, even when these come at short-term costs for their individual position in climate politics.**

2.2. Realigning climate forcing players: The role of Just Transition policies

A key finding from the climate policy literature is that the design of climate policy may sway CFAs to vote against their immediate climate interests, in the hope of making other (economic or social) gains. In this context, a rich literature on Just Transition policies has shown that, when climate policies tend to receive stronger support from the fossil fuel communities when these are associated with compensatory programs.

While a rich debate exists over what *Just Transition* policies should encompass, two overarching intervention logics can be identified – Transactional and Transformational – each rooted in different understandings of how public support for climate policy can be mobilised and sustained. These logics differ in terms of the *motivational drivers* they appeal to, the *policy targets* they prioritise, and the *type of social policies* they favour.

The Transactional logic is grounded in egotropic motivations, which commonly take the form of individuals' concern for personal economic gains. This view typically assumes a game-theoretical environment where consent to transition should be bought through targeted mitigation of material loss. In this approach, Just Transition policies are accordingly treated as a bargaining tool to compensate those directly affected by climate mitigation. Social policies are in turn mainly viewed through their ability to provide such compensation by granting concrete material gains to workers in fossil fuel industries, for example in the form such as lump-sum payment, early retirement *schemes*, or job guarantees.

In contrast, the *Transformational logic* starts from the assumption that sociotropic concerns – i.e. perceptions about the health and cohesion of one's broader community– play an equally, if not more, important role. While not every individual will experience direct economic harm from decarbonisation, many will be exposed to its secondary effects, such as the decline of community institutions or regional identity. Rather than focusing narrowly on economic losses, social policies conceived in such Just Transition packages accordingly focus less on individual compensation than on place-based strategies that restore collective capacities, with the ultimate aim of addressing the underlying structural vulnerabilities that risk exacerbating inequality and social fragmentation.

Table 1. Just transition policies – Two intervention logics

Intervention logic / Characteristics	Transactional	Transformational
Micro-drivers of change	Egotropic motivations	Sociotropic motivations
Policy target	Directly affected individu-	Indirectly affected com-

	als	munities
Social policy features	Individual (financial) compensation	Collective (social) capacitation

2.2.1. Research gap

A first contribution of this project seeks to assess the *relative role* of different types of social policy interventions included in so-called Just Transition agreements. While recent empirical work has shown that egotropic motivations only constituted one part of the story in the context of U.S. based restructuration (Gazmarian, 2024), there remains limited comparative evidence on how these logics operate in other political and institutional settings. This project therefore seeks to disentangle the transactional and transformational dimensions of Just Transition policies, examining their respective influence on political realignment behind climate-related reforms.

Today, we also know too little today about the extent to which these motivations may be mobilized by different actors to prompt a re-evaluation of the preferences of CFAs. In this study, we hypothesise that while the content of just transition policies certainly affect climate policy consent by CFAs, this effect will be further mediated by the position expressed by representative bodies holding a sufficient level of credibility in the eyes of these actors.

3. Realigning CFAs: Power Resources in the Jobs vs. Climate dilemma

3.1. The Jobs vs. Climate dilemma

Just Transition policy interventions do not happen in a vacuum. **Develop on the Jobs vs. Climate dilemma**. In the process of adoption, political actors can however influence CFAs, by shaping and defending just transition agreements that will ultimately sway them to agree to a transitional agreement.

3.2. The role of trade unions

Trade unions are a player at the crossroad of the just transition: Unions are often involved in decisions linked to closure, consolidation, or greening of climate forcing firms. Besides, unions often act as locally-embedded, political intermediaries, making them directly exposed to concerns linked to upholding the social capital of places affected by climate change.

Today, the literature is divided on the role of unions in climate change mitigation policies. Unions often fear job losses from pro-climate reforms (additional sources). However, evidence also shows that unions may consent to ambitious climate policies when these are accompanied by strong compensatory mechanisms (Kalt, 2021; Thomas & Doerflinger, 2020).

Of particular interest to this study is the fact that union's position may also affect the position of CFA regarding the adoption of climate policies. As Bolet et al. (2023) have shown, the role of unions in the negotiation and acceptance of just transition policies was critical in the phasing out of Spain coal mines.

From this comprehensive study, the authors conclude that, in contexts when union density is high, these can effectively shape workers and community preferences towards the acceptance of just transition packages. However, the channels through which just transition packages may shape preferences remain largely undefined.

In this study, we posit that the way unions change individual perceptions of climate policy interventions may occur via three different pathways:

1. First, unions may act as a *responsive player*, comforting 'climate-forcing asset holders' in their resistance to decarbonization policies.
2. Alternatively, unions may choose to act as *responsible player*, identifying the (social) conditions under which realignment of CFAs towards climate goals would be considered acceptable. In this case, we expect that these may act either by following either
 - a. an ***insider-oriented logic***, tailoring workers' support via targeted policies; or
 - b. a ***public-seeking logic***, appealing to the whole local population via community investments.

3.3. Target group: A focus on affected workers

To best distinguish between affected and non-affected populations, we target a representative sample of the main population of interest that is likely to backlash when decarbonization policies are implemented, namely: workers in CFA industries. With this focus on workers, we accordingly ask in this study:

When do workers agree to forego their current employment status for the sake of climate mitigation objectives?

4. Hypotheses

The first (baseline) aim of this project is to assess the public's relative preferences on Just Transition interventions bearing a transactional vs. a transformative intervention logic. Appeal to each of these policy designs will be assessed by testing individual preferences on trade-offs typically involved in various attributes conventionally featuring in Just Transition policies. Our first research question accordingly asks:

RQ 1: Are just transition policy tools contributing to the individual realignment of CFA workers?

4.1.1. Effect of policy design

We first posit that:

- **H1.** When climate policies are coupled with Just Transition measures, support among exposed workers increases (*H0: When climate policies are coupled with Just Transition measures, support among exposed workers does not increase.*)

We further refine these expectations by considering the specific features of Just Transition interventions, drawing on the distinction between transactional and transformative policies.

- **H1a.** *The more strongly Just Transition measures support workers, the more likely affected workers are to support transition policies.*

- **H1b.** *The more strongly Just Transition measures support affected communities, the more likely affected workers are to support transition policies.*
- **H1c.** *The effect in H1b. is weaker than the effect in H1a.*

4.1.2. Effect of union support

The second, critical, aim of this project is to assess the role played by unions in the political realignment of workers towards climate policy support.

Above, we theorised a number of mechanisms whereby which unions may affect individual inclinations in ways which could contribute to collective realignment towards CMOs (see section 3): When they act as a *responsible player*, we argued, unions can positively change perception of climate policy interventions. In the latter case, responsibility may take the form of an ‘insider-oriented’ or a ‘public-seeking logic’.

As a baseline assumption, we posit that:

- **H2.** *Endorsement of just transition policies by trade unions increases support for climate policy among workers in carbon-intensive sectors.*
(H0: Union endorsement has no effect on worker support for the policy.)

We further refine this expectation by considering the nature of union support and its interaction with the policy’s substantive features:

- **H2a.** *Union endorsement of transactional policies increases policy support among CFA workers.*
- **H2b.** *Union endorsements of transformational measures increases policy support among CFA workers.*
- **H2c.** *The effect in H2b. is weaker than the effect in H2a*

4.1.3. Heterogeneous effect linked to workers’ status

Older workers, who face reduced reemployment prospects and are more vulnerable to job loss, may evaluate Just Transition interventions differently from younger cohorts. Accordingly, we posit that the effect of Just Transition policies may vary across different groups of CFA workers, particularly based on their proximity to labour market exit:

- **H3.** *The effect of Just Transition policy interventions on support for climate policies is stronger among older workers.*
(H0: The effect of policy features does not vary by age.)

We further specify:

- **H3a.** *Older workers show stronger support for transactional measures compared to younger workers.*
- **H3b.** *In contrast, the effect of transformational measures on support is weaker among older workers compared to younger workers.*

We finally posit that older workers, who may rely more heavily on trusted intermediaries when facing uncertainty, are more responsive to union endorsement, particularly when those cues provide reassurance about compensation or job security. Accordingly, we expect

- **H3c.** *The effect of union endorsement on support for Just Transition policies is stronger among older workers.*
(H0: The effect of union endorsement does not vary by age.)

5. Empirical Strategy: A survey experiment assessing the ‘trade union effect’ in climate policy realignment of CFA

5.1. Project focus

To assess the validity of our claims, we field a survey experiment which would foresee a substantive decarbonization. For that, we focus on sectors likely to be significantly affected by decarbonization efforts: manufacturing, heavy industry (including chemicals and materials), the automobile, and the aviation sector.

Specifically, we envisage the hypothetical case of a major restructuring of national companies that would seek to reduce the company’s carbon footprint in ways that would asymmetrically impact local environments heavily relying on a CFA industry.

5.2. Population, sampling strategy and case selection

The role of unions in shaping preferences for just transition policies depends on the *strength* of unions within specific sectors (as measured by the **collective bargaining coverage**), (b) the extent of their *institutional* embeddedness (**i.e., their influence on policymaking**), and the characteristics of the targeted population.

To investigate these dimensions, we accordingly adapt our study population and sectoral focus accordingly.

We conduct a comparative study of two countries—France and Germany—where unions play distinct roles in the bargaining process.

5.3. Administration of the survey

Overall, we aim to recruit a sample of **1,500 workers** from industries likely to be impacted by the energy transition.

The survey will be administered online via the Qualtrics platform and distributed through unions representing workers in the targeted sectors. To encourage participation, the survey will include an incentive: respondents will be entered into a lottery to win a €100 voucher. This strategy enables us to reach a sample of engaged workers—those most likely to be early movers and potentially among those bearing the greatest costs in resisting decarbonization policies.

To increase participation, we aim to collaborate with trade union associations (contact was made with ETUI in Brussels) and/or the HR departments of major companies to disseminate the survey.

6. Operationalisation

6.1. The conjoint experiment

Broadly speaking our research is concerned with the following question: Under which conditions are workers willing to accept a major personal loss (in the form of current job security) in exchange for a broader collective goal (climate policy)?

In this conjoint experiment, we aim to assess which policy features of a just transition deal would lead workers in carbon-emitting industries to consent to a climate

policy package, even at the cost of a likely negative impact on their current employment status.

6.1.1. Approach

Respondents are presented with **4 pairs of hypothetical policy packages** (conjoint profiles) and asked to evaluate them.

6.1.2. Prompt

Prompt 1 - Conjoint

To address global warming, the government has adopted a five-year plan to reduce emissions in high-polluting sectors by phasing out fossil fuels.

As part of this national effort, your region has been selected for early implementation of this plan. Your company is directly affected and expects to reduce its workforce by 30% over the 5 coming years.

To limit the social costs of this transition, the government considers introducing a Just Transition package for affected workers and communities.

In the following task, you will be shown four pairs of considered Just Transition packages. Each proposal includes a combination of features such as the level of financial support, job retraining opportunities, investment in local infrastructure, and more.

For each pair, please review the proposals carefully and select the one you find more acceptable. There are no right or wrong answers—we are interested in your preferences.

6.1.3. Attributes

6.1.3.1. Financial compensation

Levels	Label	Source / real-world examples
a) No addi-	<i>You receive only standard unem-</i>	Baseline condition in many EU

tional compensation	<i>ployment benefits.</i>	settings when no deal is struck 2021 Sines (Portugal) fired power plant closure: In 2021, Portugal closed its last two coal-fired power plants—Sines and Pego—as part of its commitment to climate change mitigation. Many of the workers in Sines primarily relied on standard unemployment benefits, with limited additional compensation or support mechanisms in place.
b) Lump sum (mandatory redundancy payment or negotiated severance payment)	<i>You receive a one-time compensation equal to approximately three months of your previous salary.</i>	Poland 2024 closure of coal and lignite-fired power plants and mines: collective labor agreements envisage severance payments. Spain 2018 coal mines closure agreements: Younger miners will receive a redundancy payment of €10,000, as well as 35 days' pay for every year of service. Miners with asbestosis will receive an additional payment of €26,000. (See also: https://www.industrialunion.org/sites/default/files/uploads/documents/2018/SPAIN/spanish_plan_for_coal_eng_oct_2018.pdf)

c) Monthly income bridge	<i>You receive a monthly allowance equal to around half of your previous salary, for up to 12 months.</i>	Poland 2024 closure of coal and lignite-fired power plants and mines : Employees are entitled to one-year severance payment, which they can choose instead of the severance payments foreseen in the applicable collective labor agreements.
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6.1.3.2. Job support

Levels	Label	Source / Real-world examples
a) No additional support	<i>You have access to the standard public employment services available in your country.</i>	Baseline condition most EU countries where PES (Public Employment Services) are available to all citizens.
b) Personalized job support	<i>You receive individualised support from a personal employment advisor or career coach.</i>	Spain's approach to the coal phase-out involved individualised support for workers as part of the so-called 'Just Transition Agreements'
c) Job transition guarantee	<i>You receive a job guarantee in another sector for a limited duration period, either in the public or private sector.</i>	Transition companies in Germany

6.1.3.3. (Re-)training support

Levels	Label	Source / Real-world examples
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a) No training offered	<i>You are not offered any (re-)training program as part of the transition.</i>	Baseline condition
b) Basic training (entry-level)	<i>You receive short-term training or apprenticeship to help you secure a job in low-skilled services (e.g., logistics, care work).</i>	TBD
c) Advanced training	<i>You receive vocational or educational upskilling to help you shift to a higher-skilled profession (e.g., technician, engineer).</i>	TBD

6.1.3.4. Location of the new activity

Levels	Label	Source / Real-world examples
a) In your area: commute time under 30 minutes a day	<i>Your new activity is nearby and requires no more than 30 minutes of commuting each day.</i>	Baseline condition in most countries where relocation is voluntary and unsupported. Blankenship et al. (2022)
b) In neighbouring area: commuting time between 1 and 2 hours a day	<i>Your new activity is located in a neighbouring area within 1 to 2 hours commuting distance.</i>	TBD
c) Other region	<i>Your new activity requires you to relocate to another region.</i>	TBD

6.1.3.5. Community investment

Levels	Label	Source / Real-world examples
a) No investment	<i>No additional government investment is targeting the local community.</i>	Baseline condition for an un-managed transition.
b) Green job-creating investment	<i>The government provides additional in renewable industries to attract or retrain workers.</i>	In Asturias , in the Suroccidente JTA action area, the Institute has allocated more than 77 million euro to support business projects, municipal infrastructures and three major environmental restoration works for mining operations. In addition to the environmental restoration of degraded areas in Buseiro, Ceredo and Tormaleo, with a total outlay of 82.4 million, 10 projects have been funded for sanitation, water supply and lighting infrastructures, energy efficiency measures in industrial estates and the construction of mini-warehouses to provide facilities for companies on a temporary basis. The installation of a plant for the drying and thermal treatment of wood as a source of renewable heat has also received funding.
c) Local infrastructure upgrading	<i>The government provides additional investment to upgrade local infrastructure, such as transport or</i>	See e.g., EU Just Transition Fund projects in Poland and Eastern Germany (mobility, digital access).

	<i>broadband connectivity.</i>	
d) Welfare services investment	<i>The government provides additional investments to develop welfare services, such as hospitals or childcare infrastructure.</i>	EU JTF investments in Ireland foresee financial support to “the delivery of education, skills, and training and the associated social care infrastructure, for the purposes of supporting participation in skills training and employment”

6.1.3.6. Union position

Levels	Label	Source / Real-world examples
a) No union position	<i>The trade union representing workers in your industry has not taken a position on the government's plan.</i>	Baseline condition
b) Union opposes plan	<i>The trade union representing workers in your industry opposes the government's transition plan.</i>	In 2015, IG BCE organized a large demonstration in Berlin, with approximately 15,000 participants, protesting against the government's proposed climate protection levy, which aimed to reduce emissions from older coal-fired power plants.
c) Union supports plan	<i>The trade union supports the transition plan.</i>	TBD

6.2. The vignette experiment

To test Hypotheses 2 and 3, we implement a vignette experiment. The design follows a factorial structure in which we vary both the policy mix and the stance of the labor union. Each scenario corresponds to an ideal type of union, as outlined in the table below.

We devise the following scenario:

To fight global warming, policymakers are considering policies to move away from fossil fuels and reducing sectors that have large emissions.

Prompt 2 - Vignette

To address global warming, the government has adopted a five-year plan to reduce emissions in high-polluting sectors by phasing out fossil fuels.

As part of this national effort, your region (long reliant on coal and heavy industry) has been selected for early implementation of this plan. Your company, a large employer in the area, expects to reduce its workforce by 30% over the coming years.

To limit the social costs of this transition, the government considers introducing a Just Transition package. **FACTOR 1**

FACTOR 2

FACTOR 1 (policy design):

- Level 1 – Workers-oriented support: While the plan is still under development, it is expected to include some form of direct financial compensation and job guarantees for affected workers.
- Level 2 - Community-oriented support: While the plan is still under development, it is expected to include some form of public investment in local infrastructure, schools, and healthcare services.

FACTOR 2 (Union support) will take the following levels:

- Level 1 – Union opposition: ‘The trade union representing workers in your industry has opposed the Just Transition package.’
- Level 2 – Union support: ‘The trade union representing workers in your industry has endorsed the Just Transition package.’

7. Measurement strategy

For the conjoint experiment we use two main dependent variable as advised by Druckman and Green (2021):

- **Main outcome** – a dummy indicating the preferred policy
- (*secondary outcome for robustness check*) Respondents are asked to determine how strongly they feel about their preference from 1-3 similarly to Blankenship et al. (2022).

For the vignette experiment, we use the following dependent variables

- **Main outcome** – To what extent *do you oppose or support such a policy?* 1 = Strongly oppose, 7 = Strongly support
- **Open-ended follow up** - *Why?* Optional text entry
- **Mechanism (1)** – *How fair or unfair do you think the just-transition package is?* 1 = Very unfair, 7 = Very fair
- **Mechanism (2)** – *How legitimate or illegitimate do you think the just-transition package is?* 1 = Strongly illegitimate 7 = Strongly legitimate
- **Mechanism (3)** – *How likely or unlikely the policy will be implemented?* 1 = Very unlikely 7 = Very likely
- **Mechanism (4)** – *How likely or unlikely workers will receive compensations for the loss of their activities?* 1 = Very unlikely 7 = Very likely
- **Mechanism (5)** – *How effective or ineffective the policy is to reduce greenhouse gas emissions?* 1 = Very ineffective 7 = Very effective
- **Impact on unions support** - *How much do you support or oppose unions' actions?* 1 = Strongly oppose, 7 = Strongly support
- **Impact on government support** - *How much do you support or oppose government's actions?* 1 = Strongly oppose, 7 = Strongly support
- **Impact on unions' action** - *How likely or unlikely would you be to attend a union meeting about this transition plan?* 1 = Very unlikely, 7 = Very likely

7.1. Statistical analysis

To test Hypothesis 1, we use insights from the conjoint experiment. We estimate the Average Marginal Component Effect (AMCE) by calculating the average effect of an attribute, averaged over the joint distribution of the remaining attributes

(Hainmueller, Hopkins, and Yamamoto 2014). We cluster standard errors at the respondent level, since each subject makes multiple comparisons across which residuals are likely to be correlated.

To test Hypothesis 2, we will use both the vignette and the conjoint experiment. For the vignette, the main estimand is the Average Treatment Effect (ATE) of the Community Investment condition on support for climate policy. We estimate this quantity by (i) calculating the difference in means between the D1b and D1a groups, (ii) employing a bivariate Ordinary Least Squares (OLS) regression, and (iii) employing a multivariate OLS regression that controls for age, sex, race, education, income, and partisan identification, following the approach of Gazmararian (2024). For the conjoint, we will use the same test as in Hypothesis 1, focusing on the Community Investment attribute.

To test Hypothesis 3, we will again use both the vignette and the conjoint experiment. For the vignette, the main estimand is the Average Treatment Effect (ATE) of the Union condition on support for climate policy. We estimate this quantity by (i) calculating the difference in means between the D2b and D2a groups, (ii) employing a bivariate Ordinary Least Squares (OLS) regression, and (iii) employing a multivariate OLS regression that controls for age, sex, race, education, income, and partisan identification, again following Gazmararian (2024). For the conjoint, we will use the same test as in Hypothesis 1, focusing on the Union attribute.

7.2. Power

For the conjoint experiment, assuming an effect size of 0.03, 1,500 respondents, four tasks per respondent, and a maximum of three levels per attribute, we expect to achieve 80% power. With a larger effect size of 0.05—commonly used in power analyses on this topic (see Gazmararian 2024)—we reach 80% power with only 500 respondents. Power calculations were conducted using the following tool: <https://mblukac.shinyapps.io/conjoints-power-shiny/>.

For the vignette experiment, we benchmarked values using data from the survey experiment embedded in Gazmararian’s study. Assuming a treatment effect size of 0.1, a standard deviation of 0.5 for the outcome variable, and a power target of

80%, we estimate a required sample size of approximately 1,500 respondents (750 per group). Power calculations were conducted using the EGAP online power calculator: <https://egap.shinyapps.io/power-app/>.

8. Conclusion / expected contribution

Overall, the project may help us identify public support for stereotypical Just Transition pathways. We aim contribute to the literature on climate politics aiming at deepening our understanding of the acceptance of just transition policies. Our study echoes Colgan et al. (2021) proposition that decarbonization success depends on how and when interests mobilize. Unions may take an important role in realigning interests by actively reducing the relative power of climate-forcing assets vis-a-vis climate vulnerable assets which may ultimately participate in a flipping mechanism where individual or communities' assets shift from being dominated by climate forcing assets to climate neutral, or vulnerable assets. We bring empirical evidence and shade lights to show how unions may enable or hinder such processes.

At this stage, we could – for example, consider whether empirical evidence would support an analytical distinction between:

- ☐ **An individual compensation path**
- ☐ **A group-securisation path**
- ☐ **A collective capacitation path**

Our findings should further help us engage with the rich Just Transition literature on discussions linked to 1) desirable policy mix to facilitate the re-alignment of CFAs; 2) cognitive orientations at play in individual decision-making on Just Transition policies; 3) political drivers of / mechanisms of power resource mobilization in shaping individual realignment.

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